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Jingnan Campaign©-Using Game-Based Assessment with the Mechanism of Strategy Games for History Teaching: System Development and Learning Evaluation

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***Abstract*—Game-based assessment applied to history teaching practice is less explored in the previous studies. The purpose of the present study aims to develop a game-based history assessment system and investigate learners' technology acceptance, flow state, and learning achievement. Twenty-nine college students from northern Taiwan were recruited. The results showed that in terms of the perceived usefulness of technology acceptance model and flow state, the learners had positive feedback. With regard to learning achievement, the learners made a significant progress after using the game-based assessment system. In addition, the present study found that flow state was positively associated with learning achievement and significantly explained the variability of the learning achievement to a certain extent. With respect to gender difference, there was no significant difference in technology acceptance, flow state, and learning achievement, which indicates this game-based assessment system is not unsuitable for learners of both genders.**

Keywords—game-based assessment, scaffolding theory, flow state, technology acceptance model, learning achievement

I. INTRODUCTION

Traditional history teaching method is mainly through lecture and rote learning and uses paper and pencil tests to assess whether learners memorize the declarative knowledge, which may be harder to increase learners' motivation. However, if the game elements in game-based learning can be integrated into history teaching and assessment, guidance and feedback of game mechanisms can be taken advantage of to facilitate learning, and repetitive practice and trials can be provided, then learners' motivation and achievement will be strengthened [1]. Furthermore, game-based assessment can track learners' learning level and achievement in the context of a game [2], provide immediate feedback, and make learners master the learning contents [3]. There is, nevertheless, little empirical research focusing on history instruction using game-based assessment. Therefore, this study attempts to develop a game-based assessment system for history instruction, which is expected to improve learners' learning motivation and achievement. A game-based history assessment system, *Jingnan Campaign©*, developed by our research group (i.e., Mini Educational Game Group of National Taiwan University of Science and Technology) (http://www.ntustmeg.net), integrates a context of historical campaign with a strategy game. *Jingnan Campaign©*, through the strategy mechanism, provides historical information as intelligence clues, which are given to learners to learn the knowledge of Jingnan Campaign, a civil war in the early Ming Dynasty in China. This game is based on situated learning and takes Jingnan Campaign as the topic. By combining simulated context of the historical campaign with a strategy game, learners are able to review relevant knowledge in the game-based assessment system and make use of the intelligence clues to answer the questions. In the game, learners can command the army to fight in the battle (as shown in fig. 1) and have a test through "Jiaozhen" (叫陣) action, which is to challenge an opponent to a fight with words shouted by the general of each side when two armies meet. The contents of the words they shout are a short description about Jingnan Campaign drawn from the database of the system.

Fig. 1. Commanding the army to fight.

A part of important knowledge of the description is selected to be the options in the multiple choice items, in which players should choose one of the options to finish "Jiaozhen" (as shown in fig. 2). If a player's answer is correct, "Jiaozhen" will boost our soldiers' morale; on the other hand, if the answer is wrong, then the player will be attacked by the opponent and obtain a corrective feedback. While in "Jiaozhen", learners can analyze the intelligence clues they obtain in battles to answer the questions. The intelligence clues are designed by referring to scaffolding theory proposed by [4]. Learners are able to gain intelligence clues corresponding to the questions of "Jiaozhen" in each round, and the intelligence clues provide information for the learners to reason and analyze the answers, rather than depending on rote memory, so as to enhance the higher-order cognitive thinking. In addition to the "Jiaozhen" triggered by

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the opponent, learners are able to actively trigger "Jiaozhen" to increase the possibility to win the game and to master the history knowledge by answering the questions. If learners fail to win the game, they can repeatedly play that level to complete it. The purpose of this study aims not only to develop a game-based assessment system but also to investigate learners' technology acceptance and flow state during playing the game and to evaluate learners' learning achievement.

Fig. 2. Having a test through "Jiaozhen".

II. METHOD

Twenty-nine college students from northern Taiwan were recruited in the present study (18 males, 11 females). To measure learners' flow state and technology acceptance model, the present study adopted a flow state scale questionnaire developed by [5] and translated into Chinese and revised by [6]. The contents of the questionnaire can be divided into two dimensions, flow antecedent and flow experience. The Cronbach's alpha reliability coefficient in the current samples was 0.91. The contents of the questionnaire of technology acceptance model included two dimensions, perceived usefulness and perceived ease of use [7]. The Cronbach's alpha reliability coefficient in the current samples was 0.79. This study adopted the same test in the pre- and post-test, which was designed by two history professionals. The contents were knowledge about Jingnan Campaign in Ming Dynasty. The test item used multiple choices, the number of which was 36. The Cronbach's alpha reliability coefficient of the posttest in the current samples was 0.79.

III. RESULTS AND DISCUSSION

In terms of perceived usefulness and flow state, the mean of each dimension was above the median, 3 points, indicating that the learners thought this game was useful in learning. However, the other dimension of technology acceptance model, i.e., perceived ease of use, did not reach the median (M = 2.87), suggesting that the ease of use of the game still has room for improvement. The possibility of why the ease of use did not reach the median may be due to the learners' lack of experience of playing strategy games. With regard to understanding learners' learning achievement, a paired t-test of the pre- and post-test was conducted. The result showed that the learners made a significant progress in the history knowledge of the present study in the post-test (t = 7.72, p < .05). As for the analysis of correlation, the post-test was significantly positively correlated with flow antecedent (r = 0.39, p = .037) and with the entire flow state (r = 0.37, p = .047). Regression analysis also showed that the entire flow state significantly explained a certain amount of variance of the post test (β=0.39, p<.05), indicating that the learners' flow state during game-playing may influence learners' learning achievement. Finally, there was no significant difference in technology acceptance model, flow state, and learning achievement between genders.

IV. CONSLUSION AND SUGGESTION

The results showed that this game is beneficial to learn the relevant history knowledge. With respect to the perceived usefulness and flow state, the means were above the median. Furthermore, flow state was significantly positively correlated with the learning achievement and can explain some variance of learning achievement, suggesting flow state may affect learning achievement to a certain extent. At last, with respect to learning achievement, after using the game to learn the history knowledge, learners made a significant progress. Future research can further investigate whether learners' game experience may influence learners' ease of use of the game. In addition, learners' behavioral patterns during playing the game can be further explored [11].

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